

Global Stability vs. Directional Positivity in the 2HDM

A Structural Correction and a Coupling-Based Framework

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Abstract

A recent formal verification of the two Higgs doublet model (2HDM) has revealed an error in a widely cited stability criterion. The error originates from replacing a global boundedness condition with local directional constraints. In this paper, we show that the correct condition follows directly from quadratic completion and necessarily involves a coupling inequality between linear and quadratic contributions. We further propose a structural interpretation in which stability is understood as a resonance condition in a coupled system. This perspective naturally extends to an operator-theoretic framework.

1 Introduction

The stability of scalar potentials is a central problem in quantum field theory. In the context of the 2HDM, a widely used condition for stability has recently been shown to be incorrect via formal verification in Lean.

The issue is conceptual: a global condition has been replaced by local directional checks. This paper argues that the mistake is structurally visible and that the correct condition follows from elementary analysis.

2 Reduction to a One-Parameter Problem

The 2HDM potential can be written as a function of a scaling parameter $K_0 > 0$ and a normalized direction $\vec{k}$ with $\|\vec{k}\|^2 \leq 1$:

$$V(K_0, \vec{k}) = K_0 J_2(\vec{k}) + K_0^2 J_4(\vec{k}). \quad (1)$$

Thus, for fixed $\vec{k}$, stability reduces to a one-dimensional problem.

3 Stability as a Global Property

Definition 3.1. *The potential is stable if there exists $c \in \mathbb{R}$ such that*

$$c \leq V(K_0, \vec{k}) \quad \text{for all } K_0 > 0, \|\vec{k}\|^2 \leq 1.$$

This involves a mixed quantifier structure:

$$\exists c \forall K_0 \forall \vec{k}.$$

4 Quadratic Completion

Lemma 4.1. *For fixed $\vec{k}$, the function*

$$f(K_0) = K_0 J_2(\vec{k}) + K_0^2 J_4(\vec{k})$$

is bounded from below if and only if:

$$J_4(\vec{k}) \geq 0 \quad \text{and} \quad J_2(\vec{k})^2 \leq 4c J_4(\vec{k})$$

for some $c \geq 0$.

Proof. Completing the square:

$$f(K_0) = J_4(\vec{k}) \left(K_0 + \frac{J_2(\vec{k})}{2J_4(\vec{k})} \right)^2 - \frac{J_2(\vec{k})^2}{4J_4(\vec{k})}.$$

The minimum is finite if and only if $J_4(\vec{k}) \geq 0$, and the lower bound is controlled by the second term. $\square$

5 Failure of Directional Positivity

The incorrect claim replaces global boundedness with purely directional conditions on J_2 and J_4 . However, the dependence on K_0 cannot be eliminated.

Theorem 5.1. *Directional positivity conditions are not sufficient for global stability.*

Remark 5.2. *This has been formally demonstrated by constructing explicit counterexamples satisfying directional constraints but violating boundedness. :contentReference[oaicite:0]index=0*

6 Coupling Interpretation

The potential naturally decomposes into:

- Linear term: $J_2(\vec{k})$
- Quadratic term: $J_4(\vec{k})$

Stability requires not only positivity but a coupling condition:

$$J_2^2 \lesssim J_4.$$

7 Bamberg Model Perspective

We reinterpret the structure dynamically:

- J_2 : drift (Tri-component)
- J_4 : curvature (Okto-component)
- K_0 : scaling (energy axis)

Definition 7.1. *A system is stable if its components satisfy a resonance condition:*

$$(Tri)^2 \lesssim (Okto).$$

Thus, stability is not a geometric property of directions but a dynamical balance across scales.

8 Operator-Theoretic Outlook

The structure suggests an operator formulation:

$$H = K_0 D_1 + K_0^2 D_2, \tag{2}$$

where D_1 and D_2 encode linear and quadratic contributions.

Stability corresponds to spectral boundedness of H , which requires compatibility between D_1 and D_2 .

This aligns with Hilbert–Pólya-type ideas, where spectral properties encode global constraints.

9 Conclusion

The error identified through formalization reflects a structural issue: replacing global constraints with local checks.

We have shown that:

- Stability reduces to a quadratic minimization problem
- The correct condition is a coupling inequality
- Stability is inherently a resonance phenomenon

Key Insight: Stability is not local positivity, but global coupling.